



Day 12 Task - Cross-Source Digital Forensic Investigation



Trios cyber • 17 Sept (Edited 19 Sept)

100 points | Due 18 Sept

Practical Assignment: Cross-Source Digital Forensic Investigation

Dear Students,

For this practical assignment, you are required to conduct a small digital forensic investigation by combining evidence from your previous practical work involving Windows memory, browser activity, email/communication and mobile-device evidence.

The objective of this exercise is to understand how different forensic artifacts can be correlated to reconstruct activity, identify supporting or contradictory evidence, establish a chronological sequence and distinguish direct observations from investigative inferences.

Part 1: Define the Investigation Question

Create ONE investigation question that can be examined using evidence from your previous practical work.

Your investigation question should involve activity that can be reconstructed from multiple evidence sources.

For example:

- What sequence of user activity occurred during the selected time period?
- Can browser, communication and system activity be correlated?
- Do memory artifacts support the observed user activity?
- Do mobile and desktop communication records support each other?

Record:

- Investigation question
- Date/time period examined
- User/account context
- Evidence sources available
- Initial assumptions, if any

Do not begin with a conclusion.

Part 2: Identify the Evidence Sources



Use relevant evidence from your previous practical assignments.

Select evidence from at least THREE of the following categories:

- Windows memory
- Browser history
- Search history
- Downloads
- Email messages
- Email headers
- Attachments
- Mobile messages
- Mobile calls
- Mobile contacts
- Mobile browser activity
- Mobile application artifacts
- Photos/videos
- Location/GPS artifacts
- Backup/recovered data

For each source, record:

- Evidence source
- Artifact
- Location/path, where available
- Date/time
- User/account context
- Type of information provided
- Why it is relevant to the investigation

Take screenshots of the most important evidence.

Part 3: Examine Windows Memory Evidence

If Windows memory evidence is relevant to your investigation, review the memory evidence examined in the Windows Memory Forensics practical.

Examine available:

- Processes
- Process relationships
- Services
- Loaded modules/DLLs
- User sessions
- Network connections
- Command-line information, where available

Select at least 5 relevant memory findings where sufficient evidence is available.

For each finding, record:



- Artifact
- Process or object
- Date/time, where available
- Observed information
- Related evidence
- Interpretation

Do not classify an unusual process or module as malicious without supporting evidence.

Take screenshots of relevant Volatility 3 output.

If memory evidence is not relevant or an artifact is unavailable, state "Not Available" and explain the limitation.

Part 4: Examine Browser and Email Evidence

Review the browser and email evidence created during the earlier practical work.

Select at least:

- 3 browser artifacts
- 3 email/communication artifacts

Compare:

- Dates/times
- Websites
- Search terms
- Downloads
- Senders
- Recipients
- Subjects
- Attachments
- Message metadata
- Mailbox state

Record relationships between the artifacts.

For each relationship, record:

- Artifact 1
- Artifact 2
- Date/time
- Observed relationship
- Interpretation
- Whether the evidence supports, contradicts or does not establish the relationship

Take screenshots of the supporting artifacts.

Part 5: Examine Mobile Evidence



Review the mobile evidence created during the Mobile Device Forensics practical.

Select at least 5 relevant mobile artifacts where available.

Consider:

- Messages
- Calls
- Contacts
- Browser history
- Search activity
- Application artifacts
- Photos/videos
- Location/GPS
- Backup data
- Recovered/deleted data

Compare the mobile evidence with desktop evidence where appropriate.

For example:

- Mobile communication + desktop email
- Mobile browser activity + desktop browser activity
- Mobile media timestamp + desktop activity
- Mobile location + communication timestamp
- Mobile application activity + desktop activity

Record:

- Mobile artifact
- Related desktop artifact
- Date/time
- Observed relationship
- Interpretation

Take screenshots of important supporting evidence.

Part 6: Cross-Source Correlation

Perform a detailed correlation between at least THREE different evidence categories.

Use categories such as:

- Memory
- Browser
- Email/communication
- Mobile



Create an evidence correlation table containing:

- Evidence source
- Artifact
- Date/time
- User/account
- Observed activity
- Supporting evidence
- Contradictory evidence, if any
- Interpretation

Identify at least 5 meaningful correlations.

Also identify at least 2 areas where the evidence is incomplete, ambiguous or contradictory, where available.

Do not force a correlation when the timestamps or artifact information do not support it.

Part 7: Timeline and Activity Reconstruction

Create a chronological timeline containing at least 15 events, where sufficient information is available.

For each event, record:

- Date/time
- Evidence source
- Artifact
- Observed activity
- Interpretation
- Supporting or limiting evidence

Organize the events chronologically.

Use multiple evidence sources rather than creating the timeline from only one artifact type.

Clearly distinguish between:

- Direct evidence/observation
- Inference or interpretation

Part 8: Identify Supporting and Contradictory Evidence

Review the reconstructed activity and identify evidence that supports the proposed sequence of events.

Record at least:

- 3 supporting observations
- 2 limitations, uncertainties or contradictions, where available

For each item, record:



- Evidence
- What it supports or contradicts
- Reasoning
- Limitation

Do not remove contradictory evidence simply because it does not fit the reconstruction.

Part 9: Final Investigation Reconstruction

Based on the combined evidence, write a final reconstruction of the activity.

Your reconstruction should explain:

- What happened according to the available evidence
- When the activity occurred
- Which user/account context is associated with the activity
- Which evidence sources support the reconstruction
- Which evidence sources provide additional context
- What remains uncertain
- What cannot be established from the available evidence

Clearly distinguish between:

- Direct evidence/observation
- Inference or interpretation
- Unresolved question

Do not state an inference as a confirmed fact.

Part 10: Document Your Work

Prepare a PDF or DOCX report containing:

1. Student Information
2. Objective
3. Investigation question
4. Evidence sources
5. Windows memory evidence
6. Browser evidence
7. Email/communication evidence
8. Mobile evidence
9. Cross-source correlation
10. Supporting evidence
11. Contradictory or incomplete evidence
12. Timeline of at least 15 events
13. Screenshots
14. Direct observations vs inferences
15. Final reconstruction



16. Limitations

17. Conclusion

Each screenshot should have a short caption explaining what it demonstrates.

Submission Requirements

Submit ONE file only:

- PDF (.pdf)
- Microsoft Word document (.docx)

Important Notes

Use only evidence that you are authorized to examine.

Use the evidence created and examined during your previous practical assignments where appropriate.

Do not modify original forensic evidence.

Keep original evidence separate from working copies.

Do not treat a single artifact as proof of an entire event.

Do not ignore contradictory or missing evidence.

Use the appropriate forensic tools from the previous practical assignments when reviewing the evidence.

For Windows memory evidence, use the Volatility 3 results generated during the Windows Memory Forensics practical.

For Android evidence, use the ADB and ALEAPP results generated during the Mobile Device Forensics practical.

Your conclusion must be based on the available evidence.

Do not state an inference as a confirmed fact.

Expected Learning Outcomes

After completing this practical, you should be able to:

1. Identify relevant evidence across multiple forensic domains.
2. Correlate memory, browser, email and mobile artifacts.
3. Match timestamps and user/activity context.
4. Build a multi-source forensic timeline.
5. Identify supporting and contradictory evidence.
6. Recognize limitations and gaps in forensic evidence.
7. Distinguish direct observations from forensic inferences.
8. Construct an evidence-supported activity reconstruction.

